Audit Log Model

Software Requirements / Logical Data Model

1. Purpose

Represents a traceable record of important actions performed in the system.

2. Responsibilities

• Record relevant security-sensitive or administrative events.

• Support investigation, accountability and operational review.

• Provide event history without allowing ordinary users to alter records.

3. Proposed Logical Attributes

The following is a documentation proposal, not a claim that every field is already implemented. Confirm exact field names, types, constraints and cardinalities against the project ERD/database schema.

Attribute	Meaning / Requirement
audit_log_id	Unique event identifier.
actor_account_id	Account that initiated the action, where known.
action	Description or code for the action.
entity_type / entity_id	Affected record type and identifier, where applicable.
timestamp	Date/time the event occurred.
result_or_metadata	Outcome or limited context, excluding secrets and unnecessary sensitive data.

4. Relationships

• An audit event may reference a user account and an affected entity.

• Audit records may be generated by authentication, allotment, payment, complaint and administrative workflows.

5. Validation and Business Rules

• Audit timestamps should be consistent and reliable.

• Ordinary users must not be able to modify or delete audit events.

• Never log plaintext passwords, session secrets or other credentials.

• Access to audit logs should be restricted and retention should follow institutional policy.

6. Traceability / Notes

Audit logging is represented in both the models and utilities/services structure; define one authoritative event schema to avoid duplicate designs.

Status: Draft for review. Align this document with the approved SRS, ERD, class diagram, and institutional policies before treating it as final.